Sushant Sah

ms24btech11032@iith.ac.in | sushant.2006.sah@gmail.com | +91 9667925146 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.75
Bal Bharati Public School Pitampura, Delhi	Class XII (CBSE)	2024	91.40%
Bal Bharati Public School Pitampura, Delhi	Class X (CBSE)	2022	91.80%

SCHOLASTIC ACHIEVEMENTS

- LeetCode Top **11.34%** (1725 Contest Rating).
- **Peak rating 1638 (Expert)**, achieved in Codeforces Round 1077 (Div. 2), rank 770.
- **Rank 709 in Codeforces Round 1085 (Div. 1 + Div. 2 combined).**
- **Rank 64/[~15,000+] in Codeforces Round 1090 (Div. 4).**
- Rank 887 in Codeforces Round 1097 (Div. 2).
- Rank 476 in Codeforces Round 1074 (Div. 4).
- solved 400+ problems on leetcode
- solved 700+ problems on codeforces

SKILLS

- **Languages** - C, C++, Python, SQL, HTML, CSS, JavaScript
- **Development Tools**: Git, Github, Jupyter Notebook
- **Libraries/Frameworks**: NumPy, Matplotlib, Pandas, Scipy, React, Next.js, Express.js, Tailwind CSS.

RELEVANT COURSES

- **Programming Courses**: Introduction to Artificial Intelligence, Computational Methods in Materials Science
- **Mathematics Courses**: Calculus 1, Calculus 2, Linear Algebra, Differential Equations, Ordinary Differential Equations
- **Topics Self Studied**: Probability, OOP, OS
- **Materials Science Courses:** Semiconductor Materials, Imaging in Materials Science, Diffraction and Spectroscopy, Materials Synthesis and Characterization Lab

PROJECTS

Rubik's Cube Solver | C++ | GitHub

- Built a Rubik's Cube solver **from scratch in C++** with no external libraries; modeled the cube in 3D coordinates and encoded each cubie as a **product of primes** so that unique factorization gives every piece a collision-free ID and **O(1) position-piece lookup**.
- Implemented the full **layer-by-layer solving algorithm** (cross, first-layer corners, middle layer, last-layer orientation and permutation) in an **object-oriented class** that outputs a complete solution in **standard move notation** for any valid scramble.
- **Verified correctness over 20,000+ randomized scrambles** (100% solve rate)

Cosmic Chat | React · Node.js · Socket.IO · PostgreSQL · WebRTC | GitHub

- Built a **full-stack real-time chat application** with group chats, direct messaging, and a friend system; architected as an **SPA** with a React frontend and Express REST API backend communicating over **Socket.IO rooms** for scoped real-time broadcasts.
- Implemented **peer-to-peer video calling** using PeerJS (WebRTC) with Socket.IO for signaling; built the full call lifecycle (peer discovery, accept/decline UI, media stream setup, audio/video toggle, and cleanup) across both clients.
- Designed a **PostgreSQL schema** via Prisma ORM with a **role-based permission system** (owner / admin / member) across 6 models; managed global client state across 15+ pages using **Redux Toolkit** with cookie-based authentication.

Cellulose Group Website -- IIT Hyderabad Dept. | React · React Router · Tailwind CSS · shadcn/ui | GitHub

- Rebuilt IIT Hyderabad's Cellulose Research Group website as a multi-page React application, restructuring a single static page into 10 routed sections (Projects, Publications, People, Facility, etc.) using React Router.
- Designed a responsive navigation bar with active-route highlighting and built reusable UI components (cards, carousels, sections) with Tailwind CSS and shadcn/ui.

CERTIFICATIONS

Sampoorna Program | Coding Blocks

- Completed a structured program covering **Data Structures & Algorithms**, **Competitive Programming**, and **Full-Stack Web Development**.